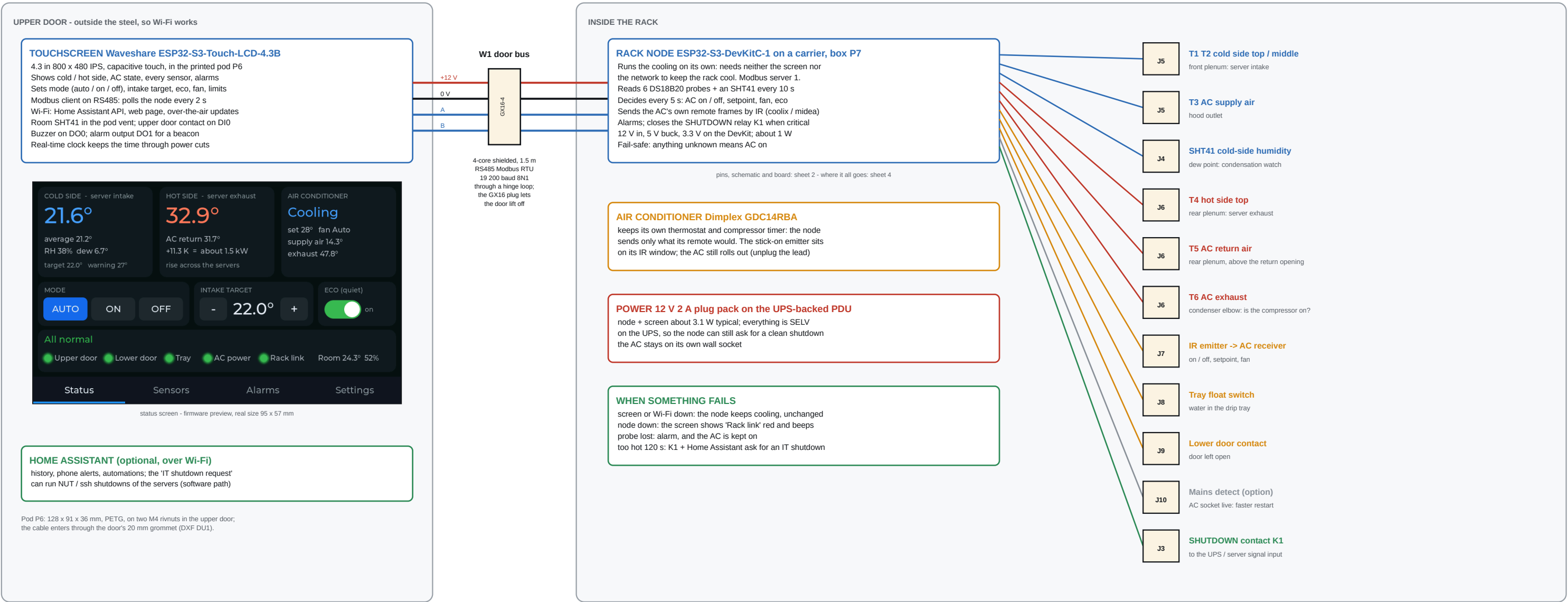
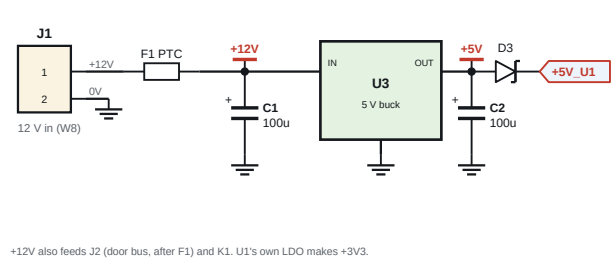


CP-SRA16-ELC-001 - System overview

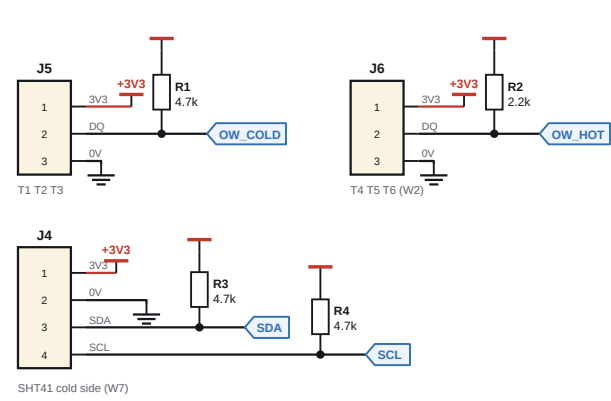


CP-SRA16-ELC-001 - Rack node carrier

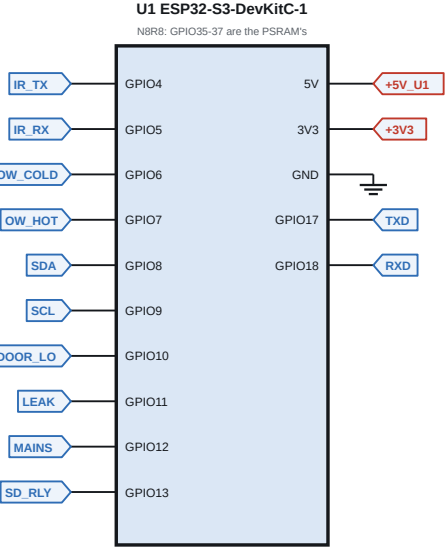
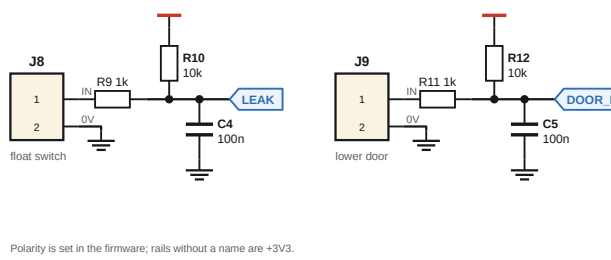
POWER



PROBES (1-Wire) + HUMIDITY (I2C)



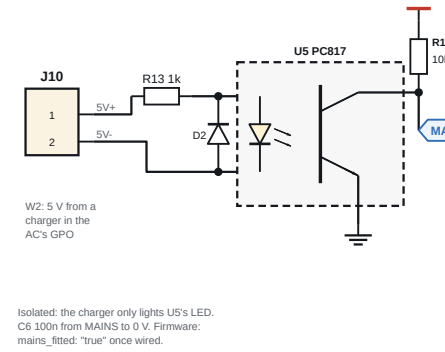
INPUTS (contact to 0 V)



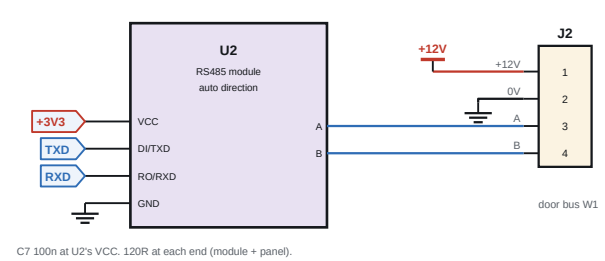
3.3 V logic: never 5 V on a GPIO

2 x 22-way sockets
USB-C ports up (lid side)
for the first flash

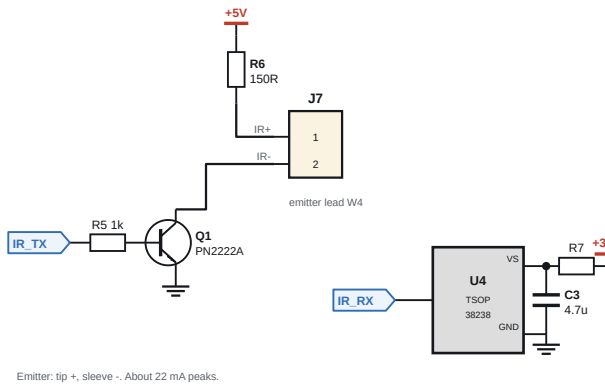
MAINS DETECT (option)



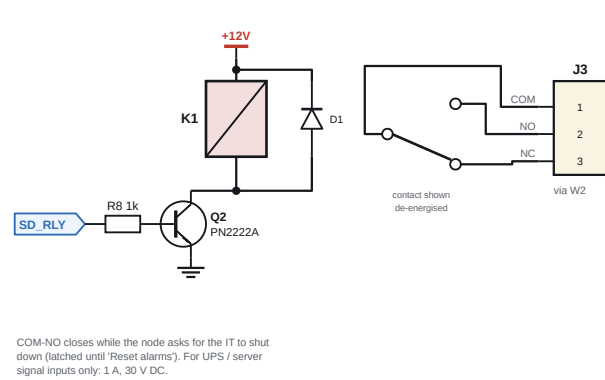
RS485 TO THE TOUCHSCREEN



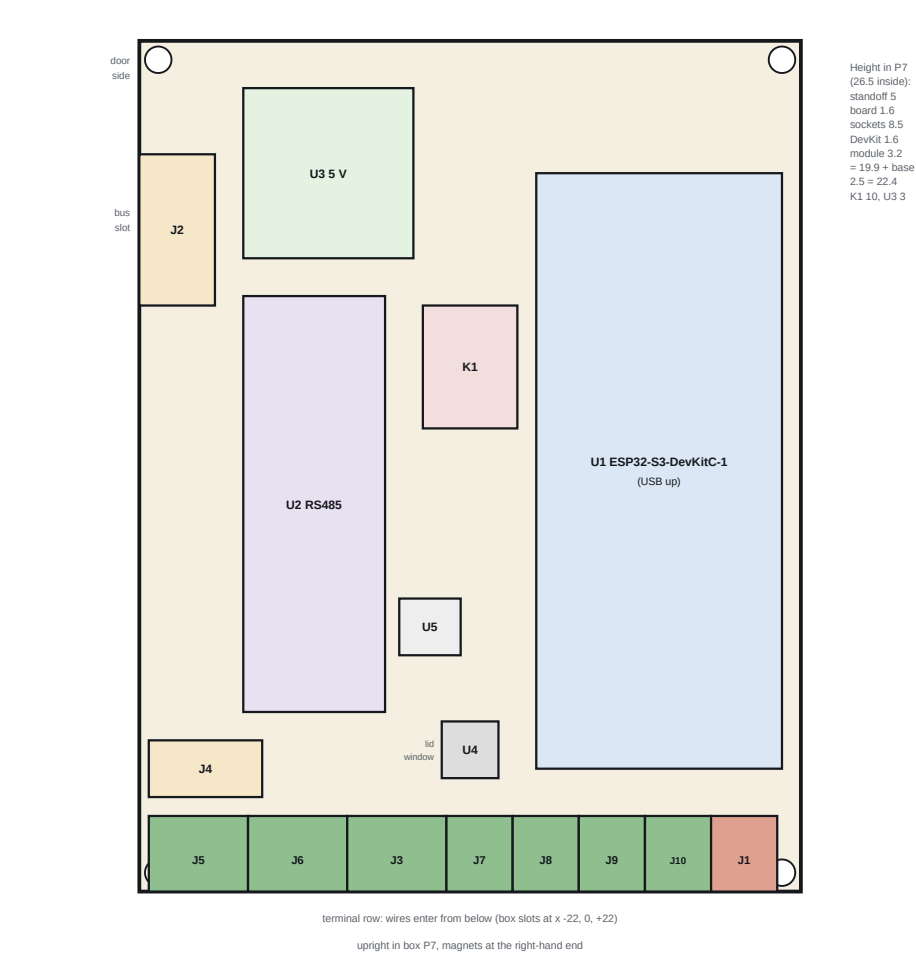
IR: EMITTER TO THE AC, RECEIVER FOR LEARNING



SHUTDOWN RELAY (dry contact)



BOARD 70 x 90, component side, 1.25 : 1



Ref	Part	Use
U1	ESP32-S3-DevKitC-1-N8R8	the controller, on 2 x 22-way sockets
U2	TTL-RS485 module, auto direction, 3.3 V	SP3485/MAX3485 class, A/B to J2
U3	5 V 1 A step-down, 7-36 V in	Pololu D24V10F5 or equal
U4	TSOP38238	IR receiver 38 kHz, under the lid window
U5	PCB17	optocoupler, mains detect
K1	Omron G5V-1-DC12	SHUTDOWN relay, SPDT 1 A 30 VDC
Q1, Q2	PN2222A	IR emitter and relay drivers
D1	1N4148	K1 flyback
D2	1N4148	U5 LED reverse guard
D3	1N5819	5 V into U1: blocks back-feed from the USB port
F1	PTC 0.5 A hold	door bus 12 V
R1	4.7k	1-Wire cold pull-up
R2	2.2k	1-Wire hot pull-up (longer bus)
R3, R4	4.7k	I2C pull-ups
R5, R8	1k	Q1, Q2 base
R6	150R 0.25 W	IR emitter current, about 22 mA peak
R7	100R	U4 supply filter
R9, R11, R13	1k	input series / U5 LED
R10, R12, R14	10k	input pull-ups
C1	100uF 25 V	12 V bulk
C2	100uF 10 V	5 V bulk
C3	4.7uF	U4 supply
C4-C7	100nF	input filters, U2 decoupling

Pin	Net	Goes to	Function
GPIO4	IR_TX	Q1 base (R5)	IR emitter, 38 kHz carrier (remote_transmitter)
GPIO5	IR_RX	U4 OUT	IR receiver: learning, protocol check (remote_receiver)
GPIO6	OW_COLD	J5-2	1-Wire, cold probes T1-T3 (R1 4.7k pull-up)
GPIO7	OW_HOT	J6-2	1-Wire, hot probes T4-T6 (R2 2.2k pull-up)
GPIO8	SDA	J4-3	I2C data, SHT41 cold-side humidity
GPIO9	SCL	J4-4	I2C clock (50 kHz on the 0.5 m lead)
GPIO10	DOOR_LO	J9-1 (R11)	lower door contact, closed when shut
GPIO11	LEAK	J8-1 (R9)	tray float switch, closes on water
GPIO12	MAINS	U5 collector	AC socket live: optocoupler pulls low
GPIO13	SD_RLY	Q2 base (R8)	SHUTDOWN relay K1 -> J3
GPIO17	TXD	U2 DI/TXD	RS485 transmit (UART1, 19200 8N1)
GPIO18	RXD	U2 RO/RXD	RS485 receive

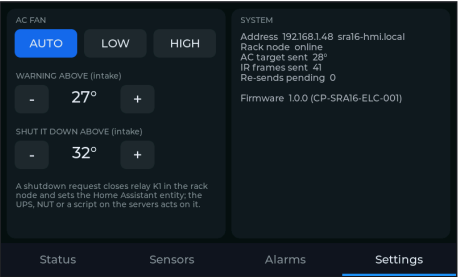
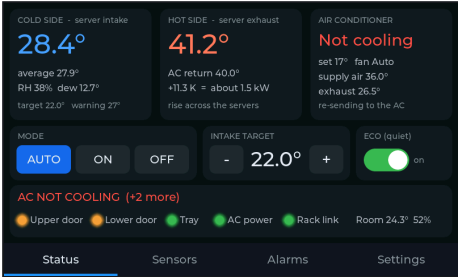
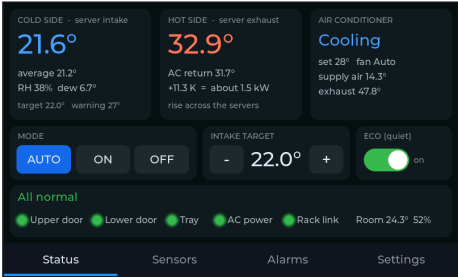
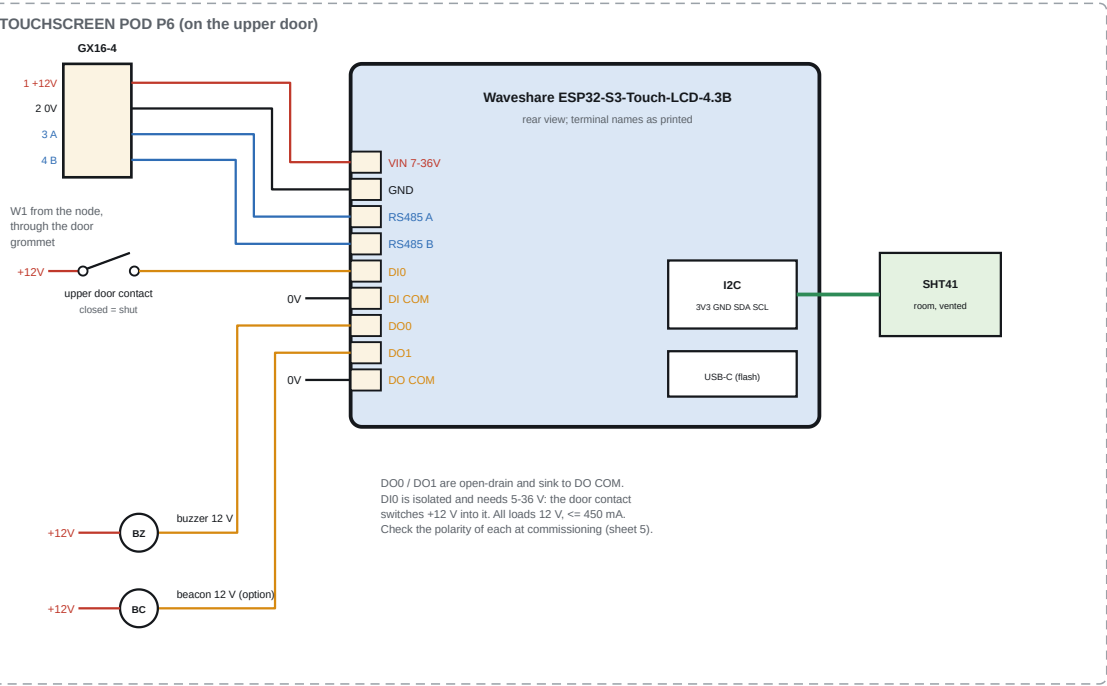
Conn	Type	Pins 1..n	To
J1	2-way terminal 3.5 mm	+12V / 0V	12 V in: plug pack on the UPS-backed PDU (W8)
J2	4-way terminal 3.5 mm	+12V / 0V / A / B	door bus W1 to the touchscreen
J3	3-way terminal 3.5 mm	COM / NO / NC	SHUTDOWN dry contact (K1) on W2
J4	JST-XH 4-way	3V3 / 0V / SDA / SCL	SHT41 cold-side humidity (W7)
J5	3-way terminal 3.5 mm	3V3 / DQ / 0V	cold probes T1 T2 T3 (W3)
J6	3-way terminal 3.5 mm	3V3 / DQ / 0V	hot probes T4 T5 T6 on harness W2
J7	2-way terminal 3.5 mm	IR+ / IR-	IR emitter socket lead W4
J8	2-way terminal 3.5 mm	LEAK / 0V	tray float switch (W5)
J9	2-way terminal 3.5 mm	DOOR / 0V	lower door contact (W6)
J10	2-way terminal 3.5 mm	5V+ / 5V-	mains detect, optional (W2)

Build: sockets for U1, then U3 and U2, then the terminal row. With U1 out, power J1: +5 V, +12 V on J2; fit U1: +3V3.

Keep the 12 V wiring on the board's door-side half; 1-Wire and I2C short and away from K1. Pull-ups and filters as drawn.

PROJECT	SRA-16 Silent AC Rack - control electronics kit
TITLE	Node carrier: schematic, board, parts
DOC NO.	CP-SRA16-ELC-001 SHEET 2 OF 5 REV A
SCALE	NTS @ A3 UNITS mm LAYOUT v1.2.0
DATE	2026-10-01 DRAWN Claude (AI) for Carbon Project
STATUS	PROTOTYPE - SELV only (12 V); no mains wiring in the kit

CP-SRA16-ELC-001 - Touchscreen pod, harness, Modbus map



W1 door bus, GX16-4

Pin	Signal	Core
1	+12V	red
2	0V	black
3	A	white / green
4	B	green / white

Shield: 0 V at the node end only. The plug sits on the cabinet side of the hinge loop: unplug it before lifting the door off.

Power budget

Load	Rail	mA	peak
Touchscreen panel (backlight on)	12 V	210	260
ESP32-S3 node incl. Wi-Fi	3.3 V (via 5 V)	90	350
Probes, SHT41, RS485, IR receiver	3.3 V	12	20
IR emitter (while sending)	5 V	0	22
Relay K1 (shutdown only)	12 V	0	13
Buzzer / beacon (alarm only)	12 V	0	30

peak 0.50 A at 12 V (plug pack 2 A), 5 V rail 392 mA peak (U3 1 A); typical 3.1 W. Plug pack 12 V 2 A, regulated, RCM approved.

HARNESS SCHEDULE (bom/electronics_harness.csv; routes on sheet 4)

W	From	To	Cable	m	Route
W1	J2 (node)	touchscreen terminals	4-core 0.22 shielded, GX16-4 inline	1.5	node box front slot - front-left corner of the front plenum - hinge loop 300 mm in spiral wrap at z 1450 (GX16 on the cabinet side) - upper door grommet - pod
W2	J6, J3, J10 (node)	T4, T5, T6; UPS contact; mains charger	8-core 0.22 shielded	3.5	bottom slot - down the front-left corner - hood port - back along the bay's left wall under the shelf (T6 on a branch tied along the exhaust duct) - up through the shelf's return opening - rear-left rail spacer (T5, then T4) - cable entry
W3	J5 (node)	T1, T2, T3	probe leads, 1 m	1.0	front-left rail spacer face, clipped
W4	J7 (node)	IR emitter (plugs in)	3.5 mm socket lead	1.5	hood port - socket tied in the bay's top-front-left corner - emitter over the AC's IR receiver
W5	J8 (node)	tray float switch	2-core 0.22	1.8	hood port - down the AC's left side gap - tray, front left
W6	J9 (node)	lower door contact	2-core 0.22	1.0	hood port - under the hood's front edge
W7	J4 (node)	SHT41 cold side	4-core 0.22	0.5	up the front-left rail spacer, P8-1 clip at z 1300
W8	J1 (node)	12 V plug pack	DC lead + 2 m extension	3.5	PDU - (rear: down the return opening, bay, hood port) - bottom slot

MODBUS MAP node = server 1, holding registers, 19 200 8N1; -32768 = no reading

Reg	Name	Type	Scale	R/W
0x00	Cold side (mean T1, T2)	S16	0.1 degC	R
0x01	Hot side (max T4, T5)	S16	0.1 degC	R
0x02	T1 cold top	S16	0.1 degC	R
0x03	T2 cold middle	S16	0.1 degC	R
0x04	T3 AC supply	S16	0.1 degC	R
0x05	T4 hot top	S16	0.1 degC	R
0x06	T5 AC return	S16	0.1 degC	R
0x07	T6 AC exhaust	S16	0.1 degC	R
0x08	Cold side RH	S16	0.1 %	R
0x09	Dew point	S16	0.1 degC	R
0x0A	Cold side average	S16	0.1 degC	R
0x0B	Rise across the servers	S16	0.1 K	R
0x0C	Setpoint sent	S16	0.1 degC	R
0x0D	State (0-8)	U16	enum	R
0x0E	Alarm bits	U16	bits 0-11	R

Flags 0x0F: b0 AC on (commanded), b1 eco, b2 shutdown request, b3 lower door open, b4 water, b5 AC socket live, b6 compressor running, b7 compressor state known, b8 mains detect fitted.

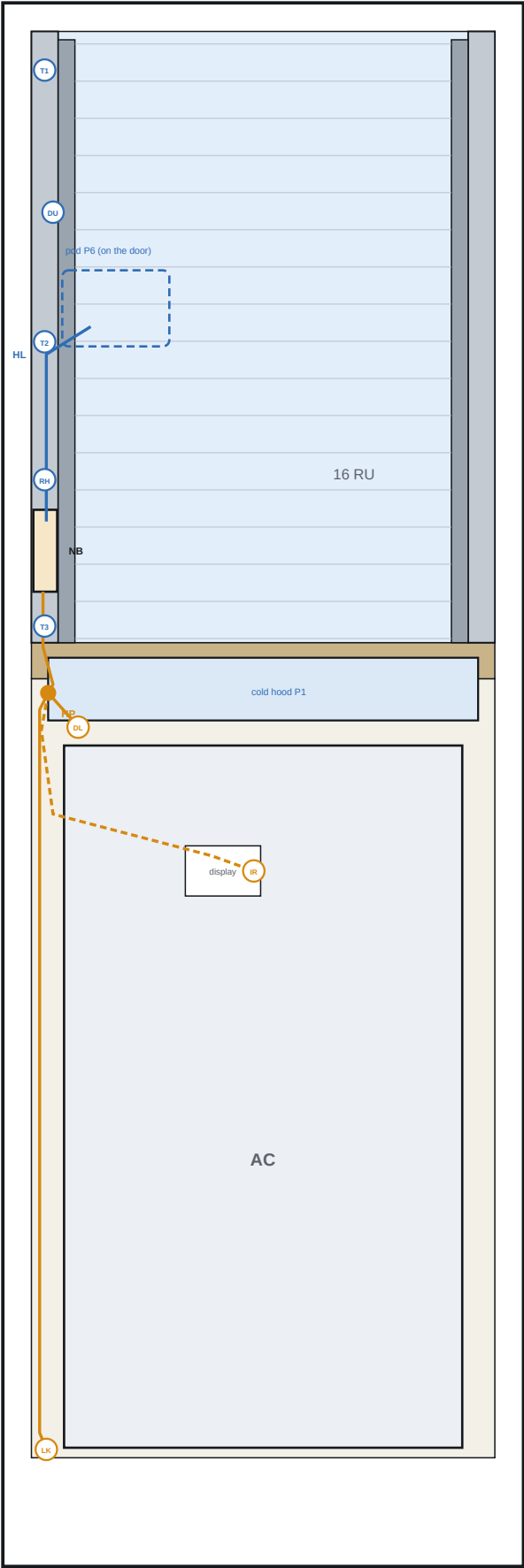
Reg	Name	Type	Scale	R/W
0x0F	Flags	U16	bits 0-8	R
0x10	Retries	U16	count	R
0x11	Uptime	U16	min	R
0x12	IR frames sent	U16	count	R
0x13	Firmware version	U16	100 = 1.0.0	R
0x20	Mode: 0 auto, 1 on, 2 off	U16	enum	R/W
0x21	Intake target	S16	0.1 degC, 18-27	R/W
0x22	Eco allowed	U16	0/1	R/W
0x23	AC fan: 0 auto, 1 low, 2 high	U16	enum	R/W
0x24	Intake warning	S16	0.1 degC, 24-35	R/W
0x25	Shutdown limit	S16	0.1 degC, 28-40	R/W
0x30	Upper door open (from the screen)	U16	0/1	W
0x31	Room temperature	S16	0.1 degC	W
0x32	Command: 1 send IR now, 2 reset alarms	U16		W
0x33	Heartbeat	U16	count	W

HOME ASSISTANT (from the screen, sra16-hmi)

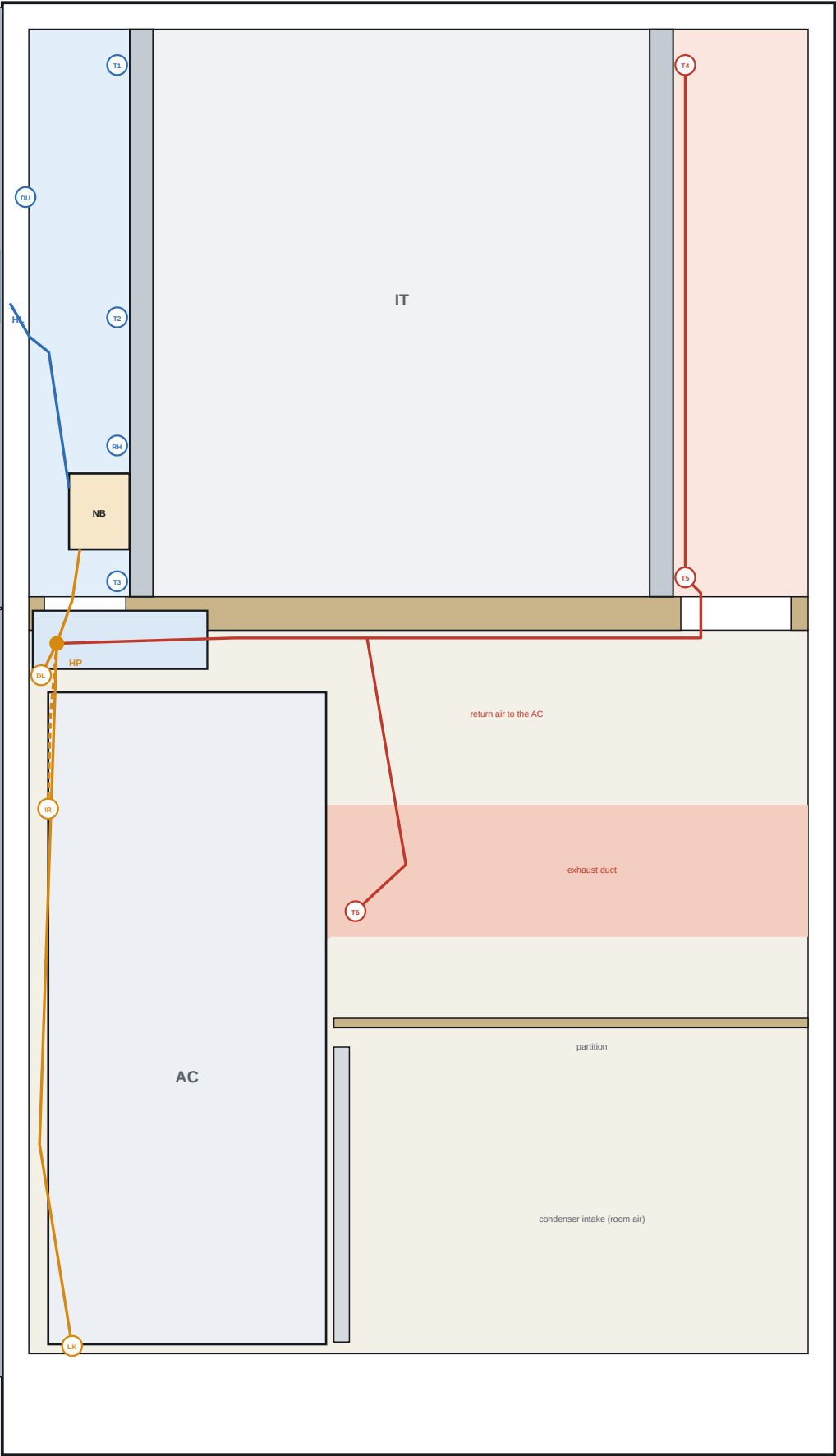
sensor	cold / hot side, T1-T6, room, RH, dew point, cold average, rise across the servers, setpoint sent, alarm bits, IR frames, uptime
binary_sensor	8 alarms (intake warning / critical, not cooling, water, door, probe, condensation, AC socket), upper door, rack link, IT shutdown request
select	cooling mode (auto / on / off), AC fan (auto / low / high)
number	intake target, intake warning, shutdown limit
switch	eco (quiet at low load)
button	re-send state to the AC, reset alarms
text_sensor	cooling state, IP address

PROJECT	SRA-16 Silent AC Rack - control electronics kit
TITLE	Touchscreen pod wiring, harness, Modbus map
DOC NO.	CP-SRA16-ELC-001 SHEET 3 OF 5 REV A
SCALE	NTS @ A3 UNITS mm LAYOUT v1.2.0
DATE	2026-10-01 DRAWN Claude (AI) for Carbon Project
STATUS	PROTOTYPE - SELV only (12 V); no mains wiring in the kit

CP-SRA16-ELC-001 - Placement in the rack, cable routes



FRONT, doors off 1:8



LEFT SECTION, looking at the left wall 1:8

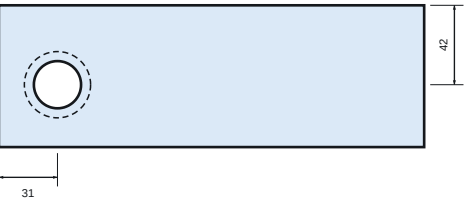
WHERE EVERYTHING GOES

Id	What	x / y / z	How
T1	Cold side top	50 / 148 / 1790	P8-1 clip on the front-left rail spacer; probe across the plenum
T2	Cold side middle	50 / 148 / 1465	P8-1 clip, same face
T3	AC supply	50 / 148 / 1125	P8-1 clip just above the shelf, tip over the hood outlet
RH	SHT41 cold side	50 / 148 / 1300	lead in a P8-1 clip, board hanging free in the air
T4	Hot side top	50 / 880 / 1790	P8-1 clip on the rear-left rail spacer, rear face
T5	AC return	50 / 880 / 1130	P8-1 clip on the rear-left rail spacer, just above the return opening
T6	AC exhaust	442 / 455 / 700	6.5 mm hole in the elbow's outer bend, probe 25 mm in, silicone
LK	Tray float switch	52 / 90 / 140	on a small alu angle bonded to the tray, front-left corner
DL	Lower door contact	90 / 50 / 1004	switch under the hood's front edge, magnet on the door lining
DU	Upper door contact	60 / 30 / 1620	switch on the door lining at the hinge side, magnet on the side lining
IR	IR emitter	300 / 59 / 832	stuck on the AC front over its IR receiver (by the display)
NB	Node box (P7)	51 / 125 / 1215	magnets on the front-left rail spacer, front face
POD	Touchscreen (P6)	135 / -30 / 1505	outside the upper door, 2 x M4 rivnuts
HP	Hood port	54 / 70 / 1045	P1-1 left end, 25 mm membrane grommet
HL	Hinge loop + GX16	40 / 50 / 1450	300 mm loop in spiral wrap, between the hinges

x from the left skin, y back from the front, z up from the floor (rack_layout v1.2.0).
Cold side (blue): front plenum. Hot side (red): rear plenum and the bay above the partition, which carries the return air down to the AC. Everything that crosses from the front plenum to the bay goes through the hood port HP; no other holes are needed. Keep probe leads 50 mm from mains cords, tied but not tensioned, and out of the condenser intake.

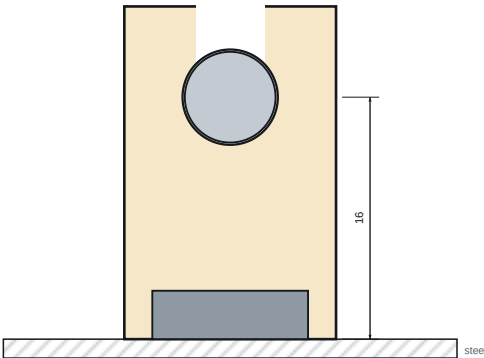
- W1 door bus; cold-side leads
- W4 IR, W5 leak, W6 lower door, via HP
- W2 rear harness: T6, T5, T4, UPS, mains

A HOOD PORT 1:4



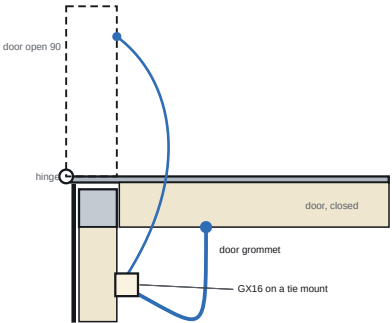
D25 through, wall thinned to 2 mm over D35 inside, 25 mm membrane grommet - pierce one hole per cable. Front face on the left.

C PROBE CLIP P8-1 2:1



6 mm probe snaps in from the top; 10 x 3 magnet glued in the base.

B HINGE LOOP, PLAN AT z 1450 1:6

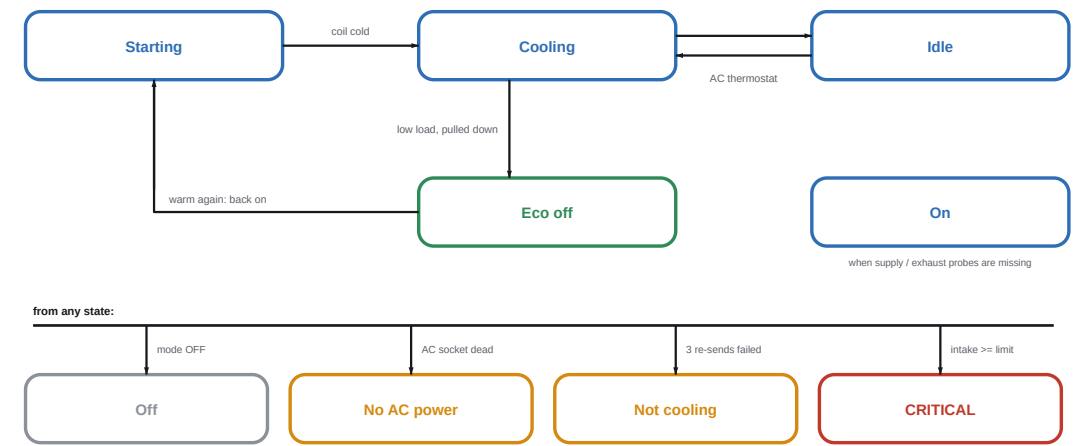


300 mm of W1 in 10 mm spiral wrap runs from the door grommet to a tie mount on the side lining, inside the seal line and half way between the hinges. Open the door fully before tying off: no tension when open, nothing pinched when shut. Thin line: the cable with the door open.

PROJECT	SRA-16 Silent AC Rack - control electronics kit
TITLE	Placement in the rack, cable routes
DOC NO.	CP-SRA16-ELC-001 SHEET 4 OF 5 REV A
SCALE	1:8 views, details as noted @ A3 UNITS mm LAYOUT v1.2.0
DATE	2026-10-01 DRAWN Claude (AI) for Carbon Project
STATUS	PROTOTYPE - SELV only (12 V); no mains wiring in the kit

CP-SRA16-ELC-001 - Control logic, alarms, commissioning

STATES (rack node, decided every 5 s)



State	Meaning
Starting	just switched on, the coil is still cooling down
Cooling	compressor running
Idle	AC on, its thermostat satisfied
On	AC on, compressor state unknown (no supply / exhaust probe)
Eco off	switched off at low load, back on when it warms
Off	switched off by the user
Not cooling	should cool, does not after 3 re-sends
No AC power	mains-detect input dead
CRITICAL	intake above the shutdown limit

Normal load: AC on; its setpoint (whole degrees) is trimmed so that the 10-minute average intake meets the target - the AC's own thermostat cycles the compressor. Low load (the servers warm the air by less than 4 K): eco pulls the rack down to target - 2, switches the AC off and back on at target + 2 or a hot side of 32. Unknown readings or over-temperature always mean AC on.

ALARMS (register 0x0E: bit)

Bit	Alarm	When	Action
0	intake warning	cold side >= warning (27)	screen, beep
1	intake critical	cold side >= limit (32)	beep; shutdown after 120 s
2	hot side warning	hot side >= 45	screen
3	AC not cooling	3 re-sends, still not cooling	beep
4	water in tray	float switch closed 5 s	beep
5	door left open	a door open 10 min	screen
6	probe missing	no reading on the cold or the hot side	screen; AC forced on
7	condensation risk	dew point 1 K over supply air for 10 min	screen
8	user OFF-overridden	off, but intake hit the warning	screen
9	IT shutdown	critical for 120 s (latched)	K1 closes, beep
10	AC socket dead	mains-detect input low	beep
11	link	no screen heartbeat 60 s / node not answering	screen

SETTINGS (defaults; screen or Home Assistant)

Setting	Default	Notes
Intake target	22.0 degC	average of the cold side the AC is trimmed to
Warning / shutdown limit	27 / 32 degC	ASHRAE A1 recommended max / hard limit
Eco	on	AC off at low load: rise across the servers < 4 K
Eco band	target -2 / +2 K	off when pulled down, on again when warm (or hot side 32)
Min on / off	15 / 5 min	compressor-friendly cycling
Setpoint range	17-30 degC	first guess target + 3, trimmed in whole degrees
Trim	20 min (2 min when 3 K hot)	on the mean error over the window
Not-cooling check	15 min (3 min when hot)	then re-send, 3 times, then alarm
Shutdown delay	120 s	cold side above the limit before K1 closes
IR send gap	>= 10 s	never floods the AC

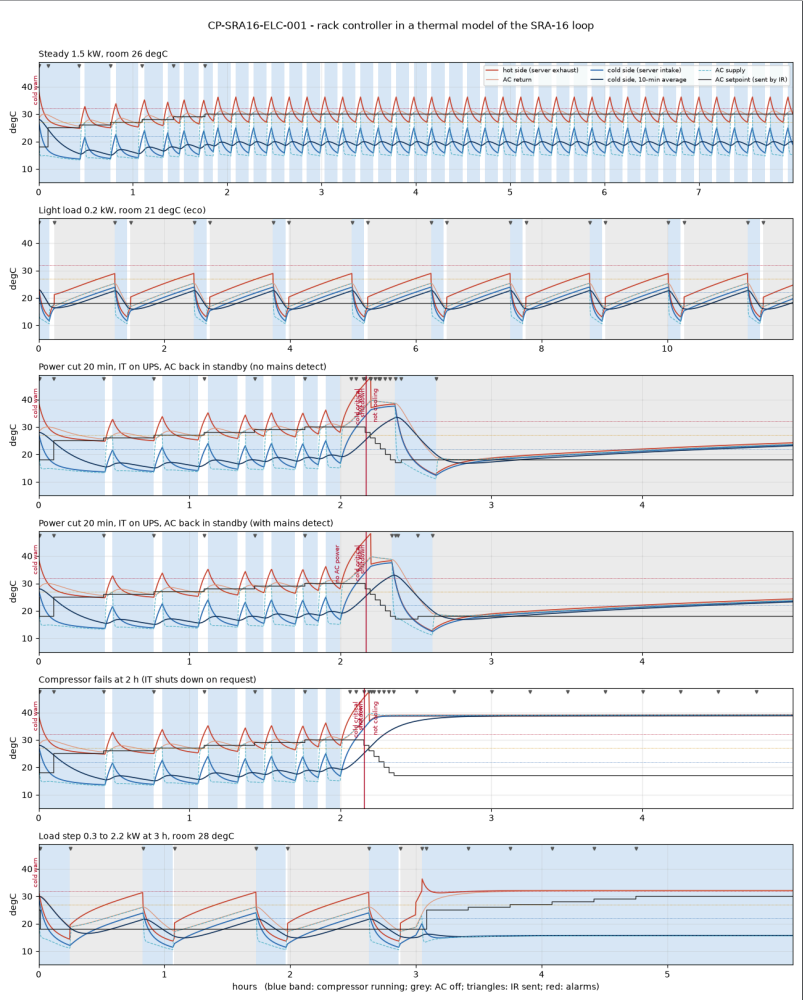
SIMULATION (the real control core in a thermal model)

Scenario	Check	Result
Steady 1.5 kW	cold side averages at or below the target	OK
	cold side peaks below the warning	OK
	no alarms once settled	OK
	setpoint steady (no hunting)	OK
Light load 0.2 kW (eco)	AC off most of the time	OK
	AC cycles, never faster than its minimum times	OK
	cold side never above the warning	OK
	AC restarted after the power returned	OK
Power cut 20 min (no mains detect)	IT shutdown requested while the AC was down	OK
	cold side peak held under 40 degC	OK
	AC restarted after the power returned	OK
	IT shutdown requested while the AC was down	OK
Power cut 20 min (with mains detect)	cold side peak held under 40 degC	OK
	not-cooling alarm within 20 min	OK
	shutdown requested	OK
	IT stopped before the cold side reached 38 degC	OK
Compressor fails at 2 h	AC on within 5 min of the step	OK
	cold side peak below the warning	OK
	AC on within 5 min of the step	OK
	cold side peak below the warning	OK

COMMISSIONING

- Bench: power the carrier from 12 V with U1 out: +5 V, +12 V on J2. Fit U1: +3V3.
- Flash both boards over USB-C (esphome run ...); 'Rack link' turns green.
- Probes: the node log lists every DS18B20 per bus at boot. Warm one at a time in your hand, put each address in sra16-node.yaml, re-flash.
- AC protocol: point the AC remote at the node box lid: the log says 'Received Coolix' (keep coolix) or 'Midea' (set midea_ir). Re-flash.
- Press 'Re-send to AC' on the Alarms screen: the AC must go to cool at the setpoint shown. Try OFF and back to AUTO. Stick the emitter where this always works.
- Doors and float: open / lift each; its dot turns amber. Backwards? add 'inverted: true' to that input.
- Unplug a probe lead: 'probe missing', 3 beeps a minute, AC kept on.
- Shutdown: set the limit to 28, warm T1 and T2: K1 closes 120 s later. Only then wire J3 to the UPS / server input. Reset with 'Reset alarms'.
- Leave it a day: the cold-side average should sit at the target.

SIMULATED DAYS (renders/controls_sim.png)



Lines: hot side (red), cold side (blue), AC supply and return. Blue band: compressor running; grey: AC off; triangles: IR frames sent; red: alarms. tools/sim_controller.py compiles sra_control.h for the PC and runs it against a lumped model of the rack and the AC - thermostat and restart delay included.

PROJECT	SRA-16 Silent AC Rack - control electronics kit
TITLE	Control logic, alarms, simulation, commissioning
DOC NO.	CP-SRA16-ELC-001 SHEET 5 OF 5 REV A
SCALE	NTS @ A3 UNITS mm LAYOUT v1.2.0
DATE	2026-10-01 DRAWN Claude (AI) for Carbon Project
STATUS	PROTOTYPE - SELV only (12 V); no mains wiring in the kit